higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

COMMUNICATION-ELECTRONICS N6

27 NOVEMBER 2019

This marking guideline consists of 11 pages.

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QUESTION 1

[Filter networks]

Constant k-section:

$$L = \frac{Z_o}{\pi . f_c}$$

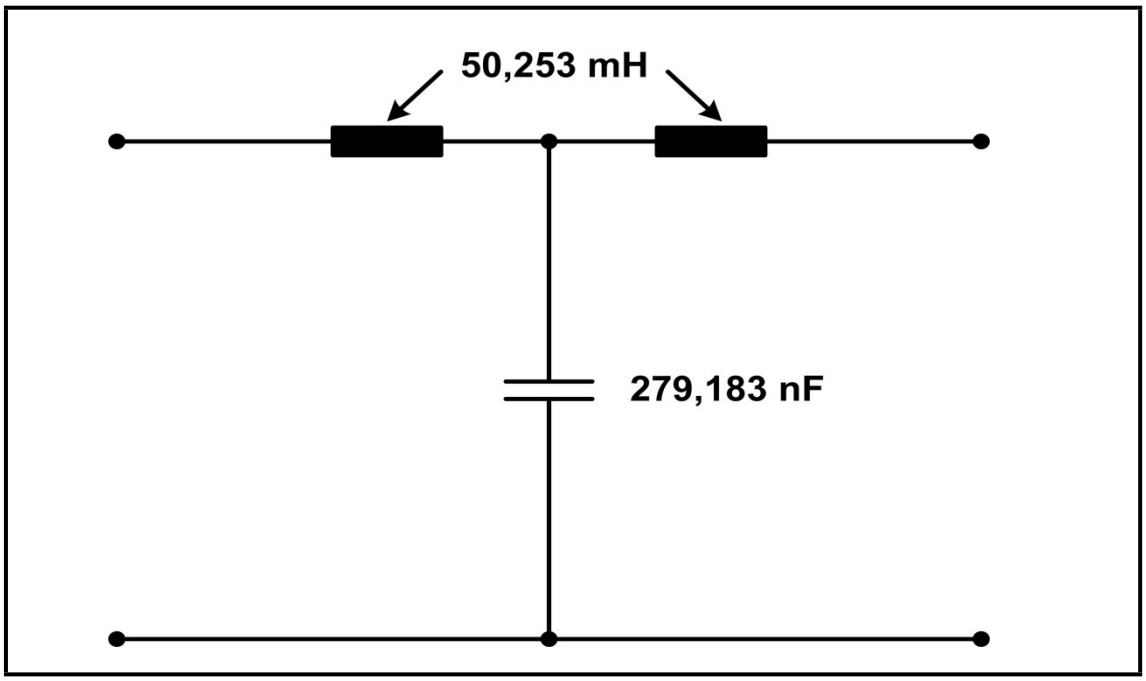
$$= \frac{600}{3,142 \times 1,9 \times 10^3}$$

= 100,506 mH

$$C = \frac{1}{\pi . Z_o . f_c}$$

$$= \frac{1}{3,142 \times 600 \times 1,9 \times 10^3}$$

= 279,183 nF



m-derived section:

$$m = \sqrt{1 - \frac{f_c^2}{f_\infty^2}}$$

$$= \sqrt{1 - \frac{(1,9)^2}{(2,1)^2}}$$

= 0,426

Shunt C = mC

$$= 0,426 \times 279,183 \times 10^{-9}$$

= 118,932 nF

$$\text{Series L} = \frac{mL}{2}$$

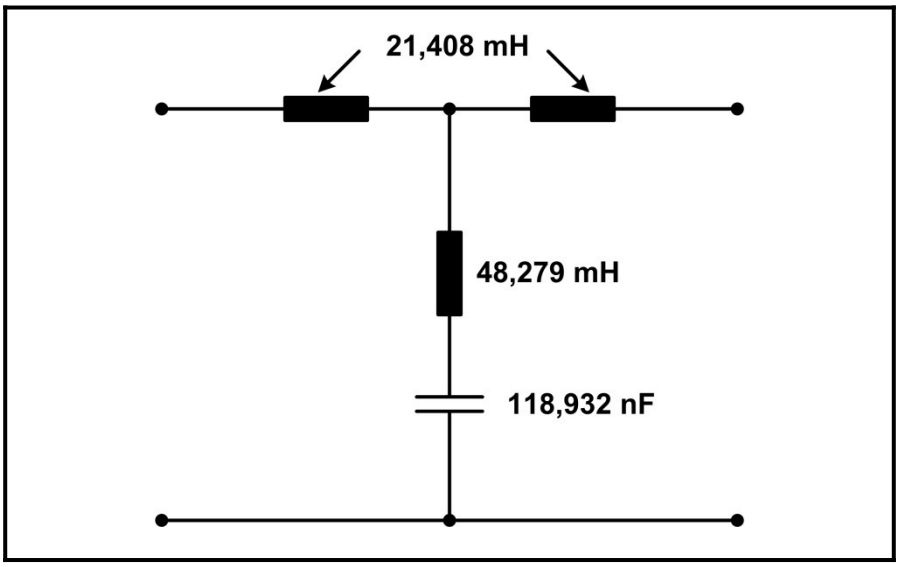
$$= \frac{0,426 \times 100,506 \times 10^{-3}}{2}$$

$$= 21,408 \text{ mH}$$

$$\text{Shunt L} = \frac{1-m^2}{4m} \times L$$

$$= \frac{1-0,426^2}{4 \times 0,426} \times 100,506 \times 10^{-3}$$

$$= 48,279 \text{ mH}$$



Half-sections (m = 0,6):

$$\text{Shunt C} = mC$$

$$= 0,6 \times 279,183 \times 10^{-9}$$

$$= 167,51 \text{ nF}$$

$$\text{Shunt L} = \frac{1-m^2}{4m} \times L$$

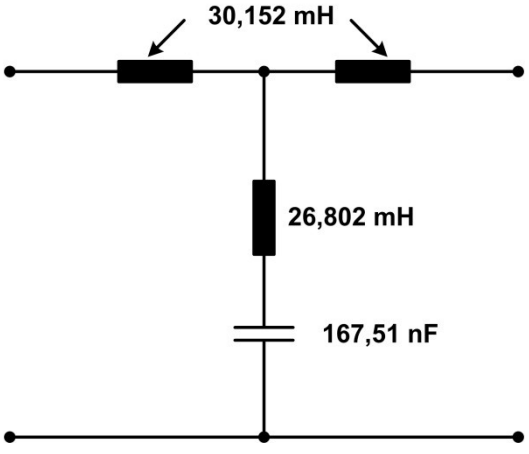
$$= \frac{1-0,6^2}{4 \times 0,6} \times 100,506 \times 10^{-3}$$

$$= 26,802 \text{ mH}$$

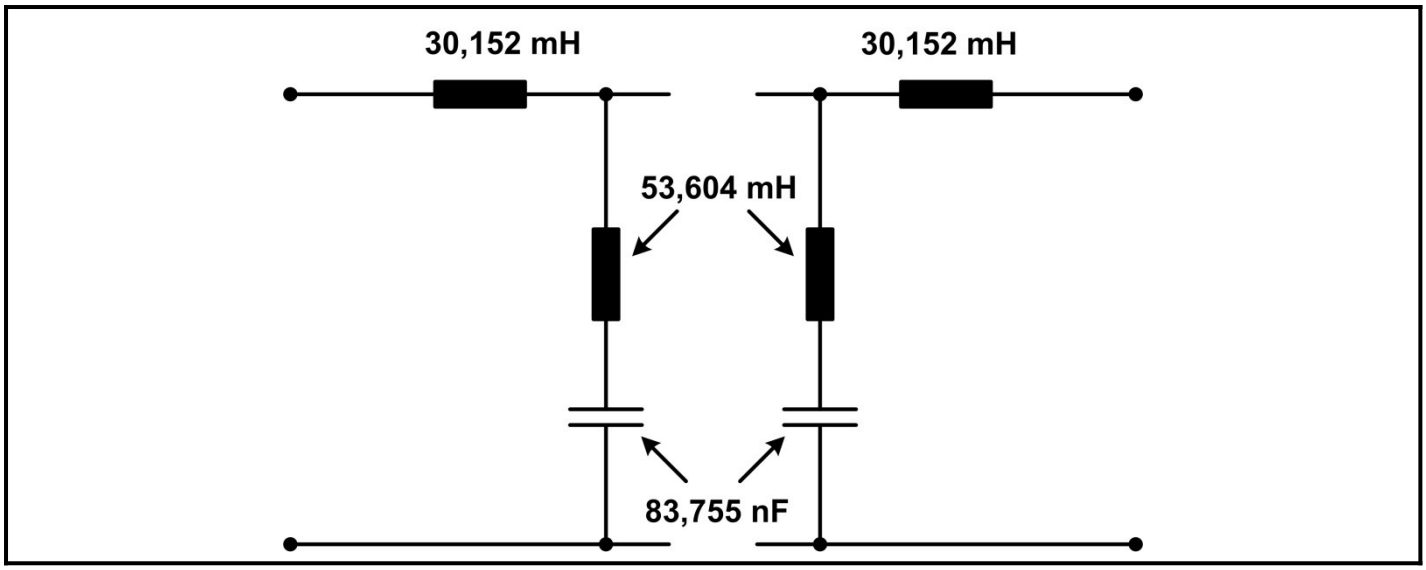
$$\text{Series L} = \frac{mL}{2}$$

$$= \frac{0,6 \times 100,506 \times 10^{-3}}{2}$$

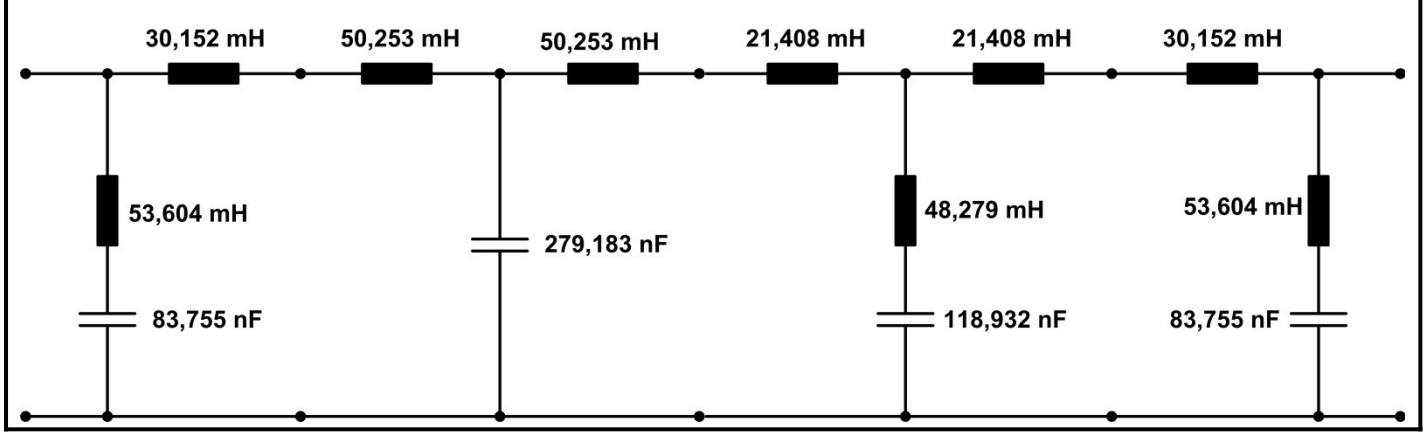
$$= 30,152 \text{ mH}$$

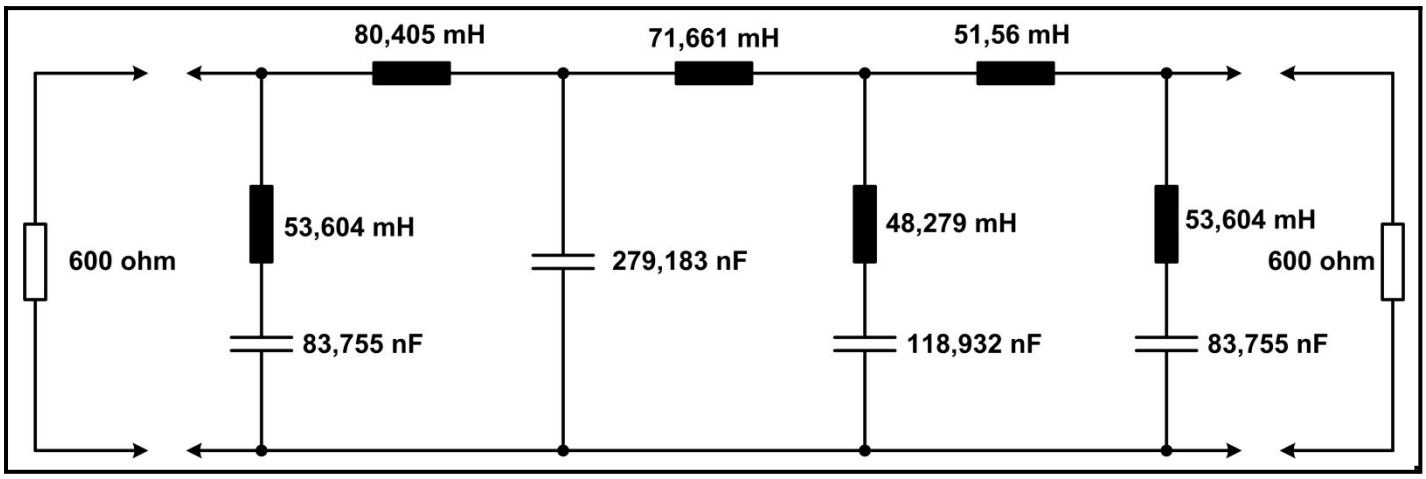


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Composite filter:





[18]

QUESTION 2

2.1 Loss = total loss – insertion loss

$\therefore$ Loss = 15 – 6

= 9 dB✓

dB = 20 log N

$\therefore$ $N = \text{antilog}\ \frac{dB}{20}$

$= \text{antilog}\ \frac{9}{20}$

= 2,818✓✓

$\therefore$ $Z_1 = R_1\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_1.R_2}\left[\frac{N}{N^2-1}\right]$

$= 560\left[\frac{2{,}818^2+1}{2{,}818^2-1}\right] - 2\sqrt{560\times 420}\left[\frac{2{,}818}{2{,}818^2-1}\right]$

= 327,571 Ω✓

$\therefore$ $Z_2 = 2\sqrt{R_1.R_2}\times\left[\frac{N}{N^2-1}\right]$

$= 2\sqrt{560\times 420}\times\left[\frac{2{,}818}{2{,}818^2-1}\right]$

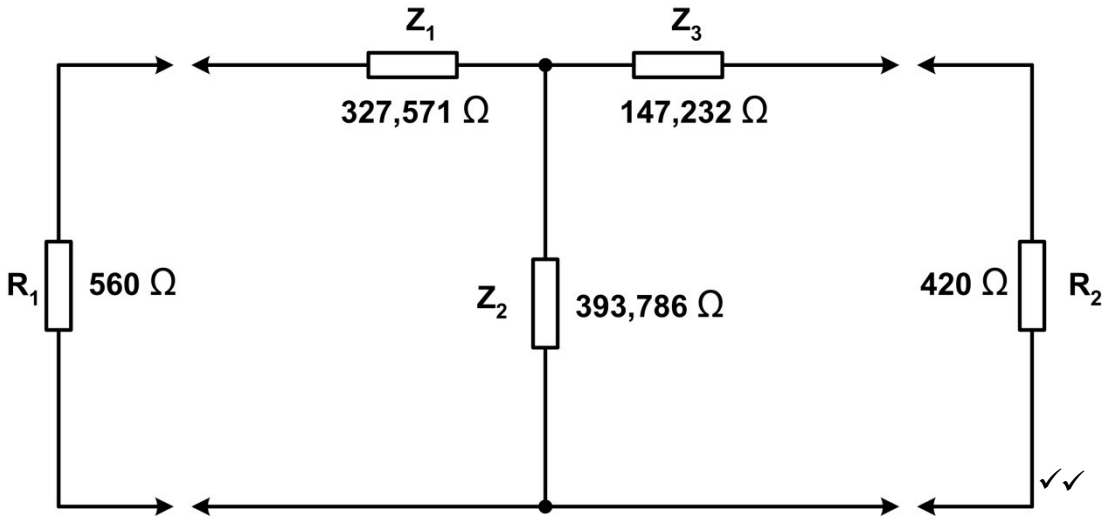
= 393,786 Ω✓

$\therefore$ $Z_3 = R_2\left[\frac{N^2+1}{N^2-1}\right] - 2\sqrt{R_1.R_2}\left[\frac{N}{N^2-1}\right]$

$= 420\left[\frac{2{,}818^2+1}{2{,}818^2-1}\right] - 2\sqrt{560\times 420}\times\left[\frac{2{,}818}{2{,}818^2-1}\right]$

= 147,232 Ω✓

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✓✓ (8)

2.2

$$Z_B = \frac{\sqrt{R_1.R_2}}{2} \cdot \left[\frac{N^2-1}{N}\right]$$

$$2\,Z_B = \sqrt{R_1.R_2} \cdot \left[\frac{N^2-1}{N}\right]$$

$$8 \times 10^3 = \sqrt{300\,.\,R_1} \cdot \left[\frac{10^2-1}{10}\right]$$

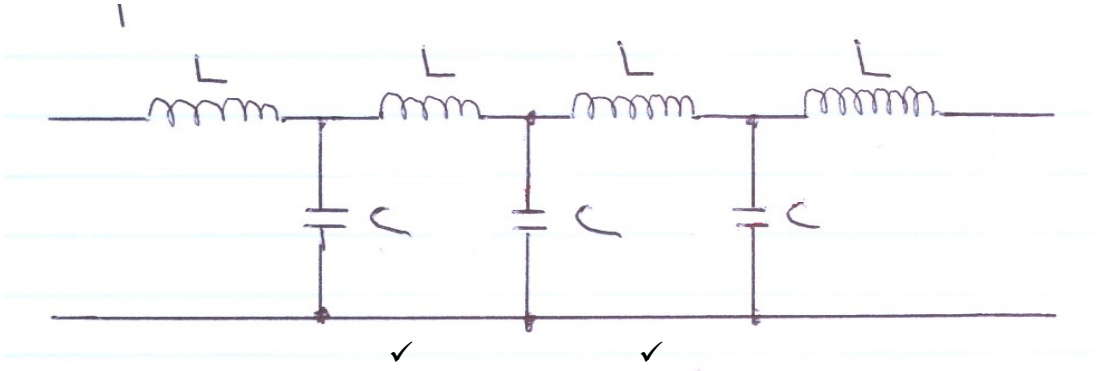
$$8 \times 10^3 = \sqrt{300\,.\,R_1} \times [9{,}9]$$

$$\sqrt{300\,.\,R_1} = 808{,}081$$

$$300R_1 = 652{,}995 \times 10^3$$

$$\therefore R_1 = 2{,}177\ k\Omega$$ (4)

2.3



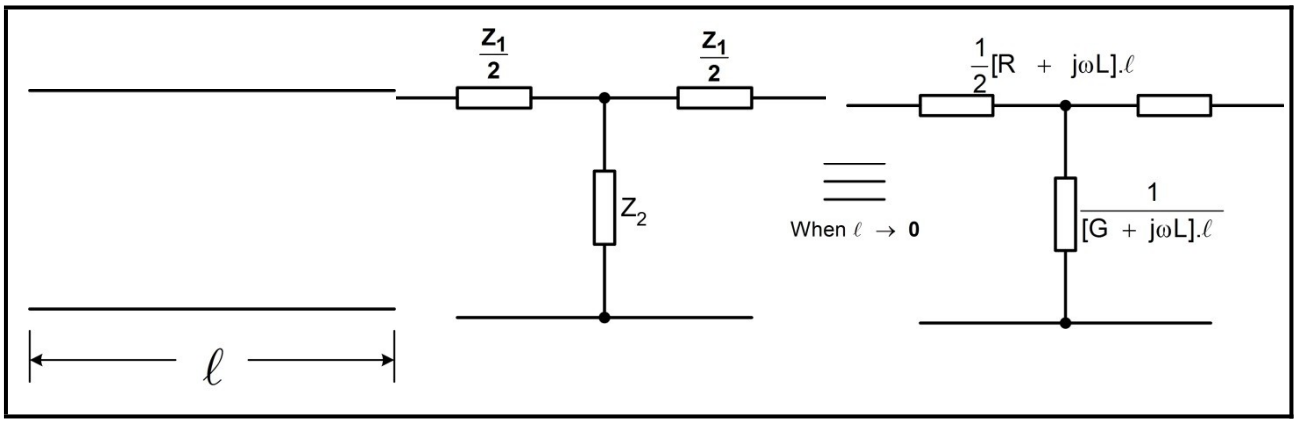
✓ ✓ (2)

2.4
- Spacing of the conductors
- Dielectric constant of the insulator
- Size of the conductors (3)

[17]

QUESTION 3

3.1



$$Z_0 = \sqrt{\frac{Z_1^2}{2} + Z_1 Z_2}$$

But $$\frac{Z_1}{2} = \frac{1}{2}[R + j\omega L] \cdot l$$

$$Z_2 = \frac{1}{[G + j\omega C] \cdot l}$$

$$\therefore \quad Z_O = \sqrt{\frac{(R + j\omega L)^2 \; l^2}{2} + \frac{(R + j\omega L) \; l}{(G + j\omega C) \; l}}$$

Let $l = 0$

$$\therefore \quad Z_O = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$ (6)

3.2 3.2.1 Tunes out reactive components

3.2.2 Measures power along a transmission line

3.2.3 Matches an unbalanced device with a balanced device

3.2.4 Determines the position and size of the first voltage maximum from the load

(4 × 1) (4)

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3.3
- Normalise impedance:

$$Z_o = \frac{R}{Z_o} \pm j\frac{X}{Z_o}$$

- Plot R on the centre line and X on the circle.
- Follow the two lines until they intersect one another and plot this point.
- Draw the circle with a radius equal to the centre of the chart and the point plotted in the previous step.
- Read off the SWR at the bottom of the circle where it intersects the centre line. (6)

3.4 3.4.1 Directional coupler
3.4.2 Slotted line

(2 × 1) (2)
[18]

QUESTION 4

4.1 4.1.1

$$En = \sqrt{4 \cdot K \cdot T \cdot \Delta f \cdot (R_1 + R_2)}$$

$$= \sqrt{4 \times 1,38 \times 10^{-23} \times 300 \times 30 \times 10^6 \times 7,8 \times 10^3}$$

$$= 62,25 \ \mu V$$ (3)

4.1.2

$$R'_{eq} = \left(R_{eq} - R_t\right) + \frac{R_{out}}{A^2}$$

$$= \left(5,6 \times 10^3 - 2,2 \times 10^3\right) + \frac{150 \times 10^3}{75^2}$$

$$= 3\ 426,667\ \Omega$$

$$F = 1 + \frac{R'_{eq}}{R_a}$$

$$= 1 + \frac{3426,667}{50}$$

$$= 69,533$$ (4)

4.2
- Solar noise
- Cosmic noise (2)

4.3 4.3.1

$$R_{eq} = R_A + \frac{R_B}{A_1^2} + \frac{R_C}{A_1^2 \cdot A_2^2}$$

$R_A = R_{1(IN)} + R_{1(eq)} R_C$ | $= R_{2(OUT)}$

$= 9 \times 10^3 + 12 \times 10^3$ | $= 30\ M\Omega$

$= 21\ k\Omega$

$$R_B = \frac{R_{1(OUT)} \cdot R_{2(IN)}}{R_{1(OUT)} + R_{2(IN)}} + R_{2(eq)}$$

$$= \frac{18 \times 10^3 \times 20 \times 10^3}{\left(18 \times 10^3\right) + \left(20 \times 10^3\right)} + 15 \times 10^3$$

$= 24{,}474\ k\Omega$

$$\therefore R_{eq} = R_A + \frac{R_B}{A_1^2} + \frac{R_C}{A_1^2 \cdot A_2^2}$$

$$= 21 \times 10^3 + \frac{24{,}474 \times 10^3}{10^2} + \frac{30 \times 10^6}{10^2 \times 15^2}$$

$= 22{,}578\ k\Omega$ (5)

4.3.2 $E_n = \sqrt{4.K.T.\Delta f.R}$

$$= \sqrt{4 \times 1{,}38 \times 10^{-23} \times 298 \times 9 \times 10^6 \times 22{,}578 \times 10^3}$$

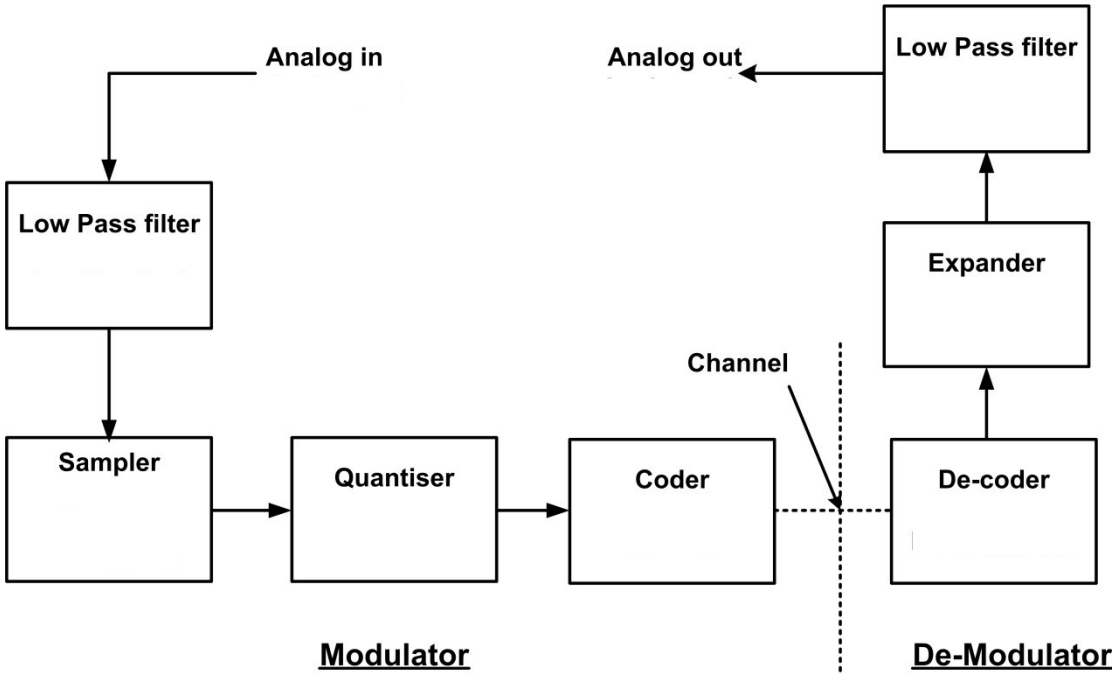
$= 57{,}815\ \mu V$ (2)

4.4 The equivalent noise resistance of an amplifier or receiver is the sum of the input terminating resistance and the equivalent noise resistance of the first stage together with the noise resistance of the previous stages referred to the input. (2)

[18]

QUESTION 5

5.1



(10)

5.2 **Advantages:**

- Better S/N characteristics for a given bandwidth
- Well suited for long-distance communications since S/N does not deteriorate
- Compatible with other digital systems

Disadvantages:

- Requires a large bandwidth
- Requires complex and expensive equipment which must be duplicated in several repeaters
- Not economical for short-distance communication (3 + 3) (6)

[16]

QUESTION 6

6.1 True (1)

6.2 6.2.1

$$P_R = Pd.A$$
$$= 1,2 \times 10^{-12} \times 12$$
$$= 14,4 \times 10^{-12} \text{ W}$$ (2)

6.2.2

$$A = \frac{G.\lambda^2}{4.\pi}$$
$$\lambda = \frac{c}{f}$$
$$\frac{A.4.\pi}{\lambda^2} \quad \lambda = \frac{3 \times 10^8}{120 \times 10^6}$$
$$= \frac{12 \times 4 \times 3,142}{2,5^2}$$
$$= 2,5 \text{ m}$$
$$= 24,131$$
$$\therefore dB = 10 \log G$$
$$= 10 \log 24,131$$
$$= 13,826 \text{ dB}$$ (4)

6.3 6.3.1 Ratio of power density radiated in a specific direction to the power density by an isotropic antenna

6.3.2 Device that will convert an electromagnetic quantity into a corresponding electrical current

6.3.3 Device that will convert an electrical current into a corresponding electromagnetic quantity

(3 × 2) (6)
[13]

TOTAL: 100

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